

Generating Blood Cells with Machine Learning: A Comparative Study of Generative Models

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Abstract. This work presents a comprehensive comparative analysis of five generative models applied to the BloodMNIST dataset for synthesizing realistic blood cell images. We implement and evaluate Variational Autoencoders (VAE), Denoising Autoencoders (DAE), Deep Convolutional GANs (DCGAN), Conditional GANs (CGAN), and Diffusion Models, systematically optimizing each through extensive hyperparameter tuning. Our evaluation using the Fréchet Inception Distance (FID) metric reveals that DCGANs achieve superior image quality (FID=37.83) despite their theoretical instability, while VAEs offer the best stability-quality trade-off (FID=67.94). The study provides practical guidelines for model selection in medical imaging applications, highlighting the importance of task-specific considerations in generative model deployment.

Keywords: Generative Models · Medical Imaging · Blood Cell Classification · VAE · DCGAN · Diffusion Models · Deep Learning

1 Introduction

Generative modeling has emerged as a fundamental paradigm in machine learning, enabling the creation of new data instances that follow the statistical distribution of training data. Unlike discriminative models that focus on learning decision boundaries between classes, generative models capture the joint probability distribution of input data, allowing for the synthesis of novel, realistic samples [1].

In the medical imaging domain, generative models offer significant potential for data augmentation, privacy-preserving data sharing, and hypothesis generation. The scarcity of labeled medical data, combined with strict privacy regulations, makes synthetic data generation particularly valuable for advancing medical AI research [2].

This work focuses on the BloodMNIST dataset, a curated collection of peripheral blood smear cell images, to evaluate the performance of different generative modeling approaches. We implement and compare five distinct model architectures: VAE, DAE, DCGAN, CGAN and Diffusion Models, each representing different philosophical approaches to generative modeling.

The main contributions of this work include: (1) a systematic comparison of generative models on medical imaging data, (2) comprehensive hyperparameter optimization for each model type, and (3) quantitative evaluation using established metrics to assess generation quality and diversity.

2 Problem Statement and Related Work

2.1 Generative Modeling Approaches

Generative models can be broadly categorized into several families, each with distinct advantages and limitations. Variational Autoencoders [3] combine the representational power of neural networks with probabilistic modeling, learning a continuous latent representation that enables smooth interpolation between data points. Generative Adversarial Networks [4] employ an adversarial training paradigm, while Diffusion Models [5] have recently gained prominence due to their training stability and high-quality sample generation.

2.2 Medical Image Generation

The application of generative models to medical imaging presents unique challenges, including the need for high fidelity in clinically relevant features, handling of class imbalances, and ensuring generated samples maintain medical validity [6]. Previous work has demonstrated the potential of various generative approaches in medical contexts, with particular success in data augmentation and anonymization tasks.

3 Methodology

3.1 Dataset Description

The BloodMNIST dataset comprises over 17,000 RGB images of peripheral blood smear cells, each measuring 28×28 pixels. The dataset includes eight distinct cell types: Neutrophils, Eosinophils, Basophils, Lymphocytes, Monocytes, Immature Granulocytes, Erythroblasts, and Platelets. For our generative modeling task, we utilize the entire dataset for training purposes, as permitted by the project guidelines.

3.2 Model Architectures

Variational Autoencoder (VAE) Our VAE implementation follows a structured approach with four main development phases:

Base Model Implementation: We implemented a conventional VAE architecture with:

- **Encoder:** 4 convolutional layers with progressive dimensionality reduction ($28 \times 28 \rightarrow 14 \times 14 \rightarrow 7 \times 7 \rightarrow 4 \times 4$), Batch Normalization and LeakyReLU after each layer
- **Decoder:** 4 transposed convolutional layers with progressive expansion ($4 \times 4 \rightarrow 7 \times 7 \rightarrow 14 \times 14 \rightarrow 28 \times 28$), Batch Normalization and LeakyReLU, final Tanh activation
- **Loss Function:** Combined MSE (70%) + L1 (30%) reconstruction loss with KL divergence regularization ($\beta=0.5$)
- **Training:** Adam optimizer (lr=3e-4), 100 epochs, periodic FID evaluation

The base model achieved an initial FID score of 116.30 ± 0.56 .

Manual Improvements: We enhanced the base model through several architectural modifications:

- Increased latent dimension from 128 to 256
- Deepened encoder with increased channels ($64 \rightarrow 128 \rightarrow 256 \rightarrow 512$)
- Implemented LeakyReLU ($\alpha=0.2$) activations
- Adjusted loss weights (70% MSE + 30% L1 reconstruction, $\beta=0.3$ KL weight)
- Added CosineAnnealingLR learning rate scheduling (min lr=1e-5)

These improvements reduced the FID score to 102.48 ± 0.13 .

Hyperparameter Optimization: We conducted systematic grid search over key parameters (Table 1):

Table 1: VAE Hyperparameter Search Space and Optimal Configuration

Parameter	Search Space	Optimal Value
Latent dimensions	[64, 128]	64
KL weights	[0.2, 0.4, 0.6]	0.2
Learning rates	[2e-4, 3e-4, 5e-4]	5e-4
Batch sizes	[128, 256]	128

The best configuration (20 epochs training) achieved FID=122.22 with the optimal parameters shown above.

Final Training: The optimized model underwent complete training (100 epochs), achieving a final FID score of 67.94 ± 0.37 .

Denoising Autoencoder (DAE) Our DAE implementation explores two distinct approaches:

Reconstruction-focused DAE: This represents the traditional application of denoising autoencoders, where the model learns to reconstruct clean images from noisy inputs. We add Gaussian noise ($\sigma = 0.2$) to input images and train the model to recover the original clean versions.

Generation-focused DAE: As an experimental approach, we investigate the potential of using trained DAE architectures for direct image generation by feeding pure Gaussian noise through the decoder network.

The DAE architecture employs a convolutional encoder-decoder structure with batch normalization and combines multiple loss components:

$$\mathcal{L}_{DAE} = 0.5 \cdot \mathcal{L}_{MSE} + 0.3 \cdot \mathcal{L}_{L1} + 0.2 \cdot (1 - SSIM) \quad (1)$$

Deep Convolutional GAN Our DCGAN followed a three-phase approach.

Base Model Implementation:

- **Generator:** 5 transposed convolutional layers expand noise input ($z \in \mathbb{R}^{100}$) from 1×1 to 64×64 resolution. Each layer uses BatchNorm and ReLU, with a final Tanh activation (scales output to $[-1, 1]$).
- **Discriminator:** 5 convolutional layers reduce 64×64 RGB images to a scalar. Spectral Normalization and LeakyReLU (slope=0.2) are used throughout. Final layer omits Sigmoid due to BCEWithLogitsLoss.
- **Loss:** Binary Cross Entropy with Logits. Regularization includes real label smoothing (0.9–1.0), 5% label flipping, and extra generator updates if discriminator loss is lower than 0.01.
- **Training:** Adam optimizer (lr=1e-4, betas=(0.0, 0.9)), 150 epochs.

The best-performing model achieved a minimum FID score of 42.30 at epoch 150.

Table 2: DCGAN Hyperparameter Search Space and Optimal Configuration

Parameter	Search Space	Optimal Value
Latent dimension	[100, 128]	100
Feature maps	[64, 128]	128
Generator LR	[5e-5, 1e-4]	1e-4
Discriminator LR	[5e-5, 1e-4]	1e-4
Batch size	[128, 256]	128

Hyperparameter Optimization: The grid search covered these parameters (Table 2):

The optimal configuration achieved FID=107.34 in preliminary testing.

Final Training: The optimized model underwent complete training (150 epochs), achieving a final FID score of 37.83 ± 0.49 .

Conditional GAN Like our DCGAN, our Conditional GAN also followed a three-phase approach.

Base Model Implementation:

- **Generator:** 5 transposed convolutional layers upsample a concatenated noise vector and label embedding from 1×1 to 64×64 . BatchNorm and ReLU follow each layer, with Tanh at the output to scale to $[-1, 1]$.
- **Discriminator:** Receives a 64×64 RGB image and a label embedding reshaped to $1 \times 64 \times 64$, concatenated along channels. Uses 5 convolutional layers with Spectral Normalization and LeakyReLU (slope = 0.2).
- **Loss:** Binary Cross Entropy with Logits. Regularization includes label smoothing (real $[0.9, 1.0]$), 5% label flipping, and extra generator updates when discriminator loss is lower than 0.01.
- **Training:** Adam optimizer (lr = 10^{-4} , betas = (0.0, 0.999)), 150 epochs.

The best-performing model achieved a minimum FID score of 247.71 at epoch 150.

Hyperparameter Optimization: The search space and optimal values are shown in Table 3:

Table 3: CGAN Hyperparameter Search Space and Optimal Configuration

Parameter	Search Space	Optimal Value
Latent dimension	[100, 128]	100
Feature maps	[64, 128]	128
Generator LR	[5e-5, 1e-4]	1e-4
Discriminator LR	[5e-5, 1e-4]	1e-4
Batch size	[128, 256]	128
Embedding size	[50, 100]	100

The best configuration achieved FID=280.33 in limited training.

Final Training: The optimized model underwent complete training (150 epochs), in which achieved a final FID score of 146.99 ± 0.83 .

Diffusion Model Our diffusion model implementation followed a similar four-phase approach:

Base Model Implementation: Initial implementation included:

- **U-Net Architecture:** 3 downsampling layers ($64 \rightarrow 128 \rightarrow 256$ channels), bottleneck layer (256 channels), 3 upsampling layers with residual connections
- **Diffusion Process:** Cosine noise scaling (1000 steps), MSE loss between predicted and actual noise
- **Training:** Adam optimizer (lr=2e-4), batch size 128, 100 epochs
- **Sampling:** DDIM accelerated sampling for generation

The base diffusion model achieved an initial FID score of 134.26 ± 1.30 .

Manual Improvements: We enhanced the architecture with:

- Attention blocks in the bottleneck for long-range dependencies
- Improved GroupNorm (8 groups) for training stability
- Enhanced residual connections with automatic channel adjustment
- CosineAnnealingLR scheduler with weight decay ($1e-4$)
- Gradient clipping at 1.0 for stability
- Increased training to 150 epochs

These improvements reduced the FID score to 113.72 ± 0.77 .

Hyperparameter Optimization: The search parameters and optimal values are shown in Table 4:

Table 4: Diffusion Model Hyperparameter Search Space and Optimal Configuration

Parameter	Search Space	Optimal Value
Learning rates	[$1e-4$, $2e-4$]	$2e-4$
Batch sizes	[64, 128]	64
Base channels	[32, 64]	64
Dropout rates	[0.0, 0.1]	0.1
Attention usage	[True, False]	True

The best configuration achieved FID=107.34 with these parameters.

Final Training: The optimized model underwent complete training (150 epochs), achieving a final FID score of 73.43 ± 0.11 .

3.3 Experimental Setup

Training Configuration All models are trained using consistent hardware configurations and evaluation protocols. We employ standard optimization techniques including Adam optimizer with appropriate learning rate scheduling and early stopping mechanisms based on validation loss convergence.

Evaluation Methodology Following the project specifications, we evaluate each model using the Fréchet Inception Distance (FID) metric. For each model, we:

1. Sample 10,000 real images randomly from the complete dataset
2. Generate 10,000 synthetic samples using the trained model
3. Compute FID scores using pre-trained InceptionV3 features
4. Repeat the evaluation across 5 independent runs with different random seeds
5. Report average FID scores with standard deviations

4 Results and Analysis

4.1 Variational Autoencoder Results

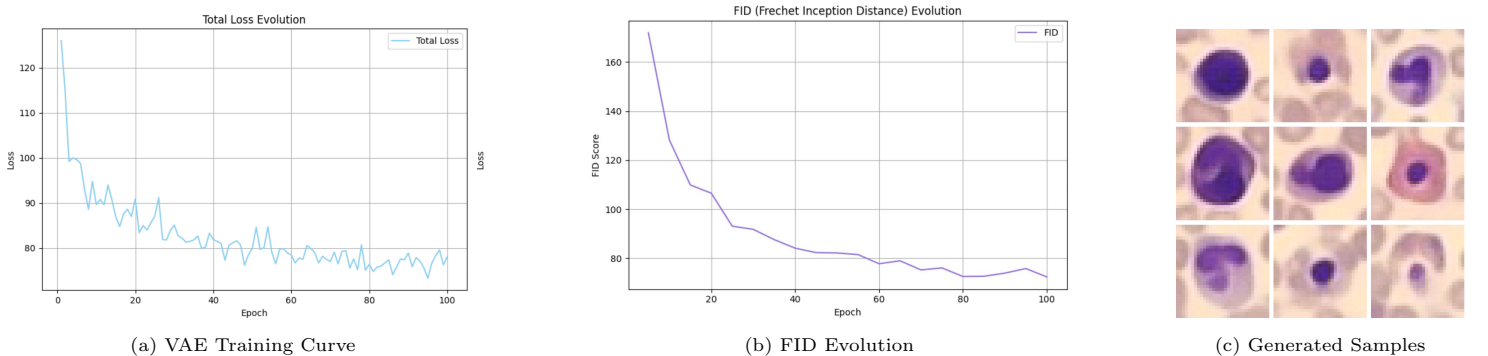


Fig. 1: VAE Training Results: (a) Training and validation loss curves showing model convergence, (b) FID score evolution throughout training showing the model’s generative performance improvement, (c) Examples of generated blood cell images demonstrating the model’s ability to capture diverse cell morphologies.

Our VAE implementation demonstrated progressive improvement through each development phase:

- **Base Model:** Achieved $\text{FID}=116.30 \pm 0.56$ with conventional architecture
- **Manual Improvements:** Reduced FID to 102.48 ± 0.13 through architectural enhancements
- **Grid Search:** Identified optimal hyperparameters ($\text{FID}=122.22$ in limited 20-epoch training)
- **Final Model:** Achieved best $\text{FID}=67.94 \pm 0.37$ with full training of optimized configuration

Table 5: VAE Performance Results

Configuration	Learning Rate	Latent Dim	FID Score	Std Dev
Base Model	3e-4	128	116.30	0.56
Manual Opt.	5e-4	256	102.48	0.13
Grid Search	5e-4	64	122.22	-
Final Model	5e-4	64	67.94	0.37

4.2 Denoising Autoencoder Results

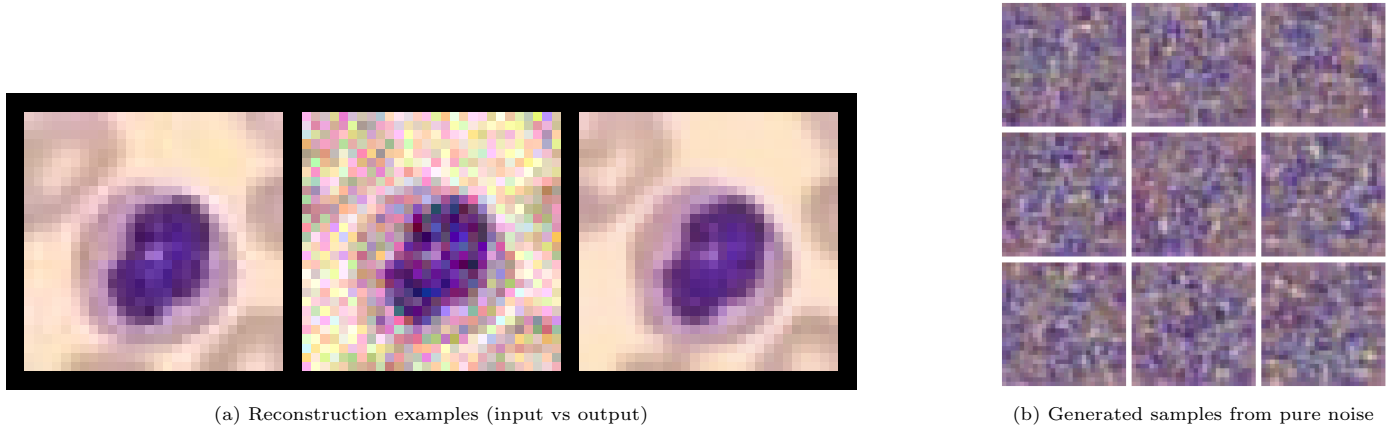


Fig. 2: DAE Performance: (a) Excellent reconstruction capability demonstrated by side-by-side comparisons. (b) Poor generation quality when attempting to create images from random noise.

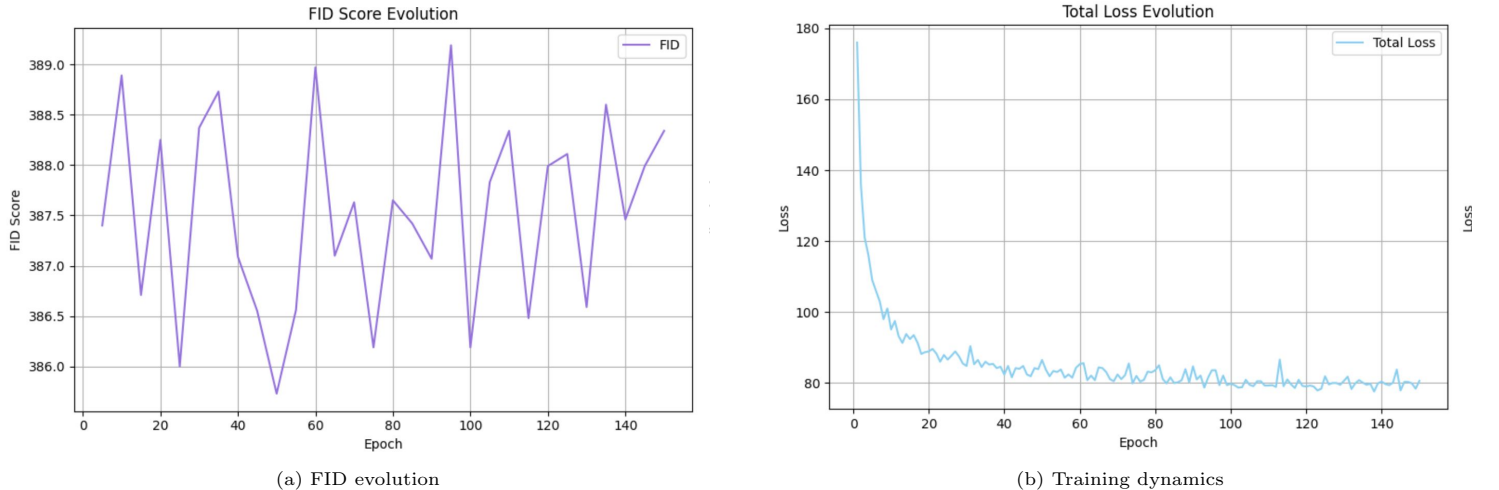


Fig. 3: Training behavior of generation experiment: (a) Consistently high FID scores (384.32 ± 0.30) confirm DAEs' fundamental limitation in image generation. (b) Reconstruction loss shows normal convergence ($176 \rightarrow 80$) with minor fluctuations (± 5 units), demonstrating effective denoising capability despite poor generative performance.

Our DAE implementation revealed a clear dichotomy:

- **Reconstruction:** Achieved excellent $\text{FID}=9.67 \pm 0.05$ for denoising task with noise level of $\sigma = 0.2$
- **Generation:** Performed poorly with $\text{FID}=384.32 \pm 0.30$, confirming theoretical limitations

4.3 Deep Convolutional GAN Results

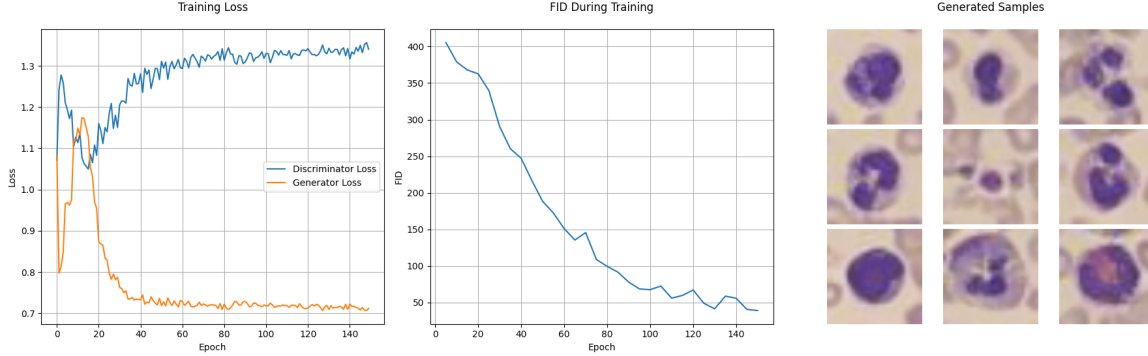


Fig. 4: DCGAN Training curve, FID during training and some generated samples: (Left & Middle) Training loss and FID curves show model stabilization and great performance. (Right) Examples of generated blood cell images demonstrating the model’s ability to capture diverse cell morphologies.

With each development phase in the DCGAN, a great improvement was observed.

Table 6: DCGAN Performance Results

Configuration	Learning Rate	Latent Dim	Feature Maps	FID Score	Std Dev
Base Model	1e-4	100	128	41.17	0.20
Grid Search	1e-4	100	128	269.60	-
Final Model	1e-4	100	128	37.83	0.49

- **Base Model:** Achieved $\text{FID}=41.17 \pm 0.20$ with conventional architecture
- **Grid Search:** Identified optimal hyperparameters ($\text{FID}=269.60$ in limited 25-epoch training)
- **Final Model:** Achieved best $\text{FID}=37.83 \pm 0.49$ after training with the best configuration of hyperparameters

4.4 Conditional GAN Results

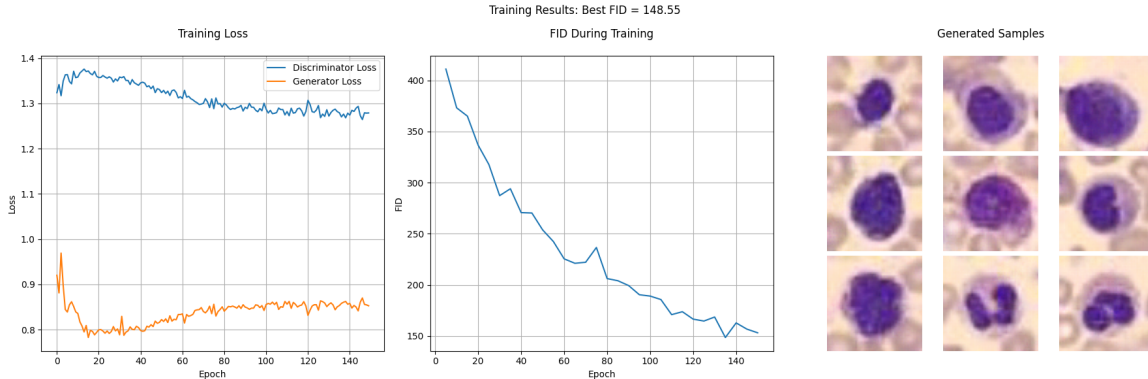


Fig. 5: Conditional GAN Training curve, FID during training and some generated samples: (Left & Middle) Training loss and FID curves show model stabilization and great performance. (Right) Examples of generated blood cell images demonstrating the model’s ability to capture diverse cell morphologies.

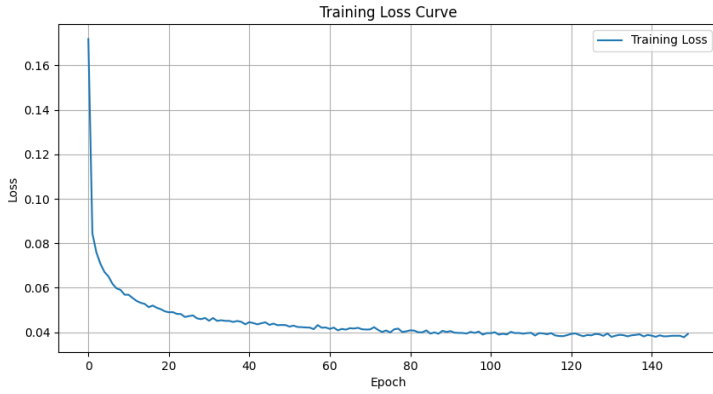
Each development phase of the Conditional GAN led to modest improvements, but the FID scores remained higher compared to other models.

Table 7: Conditional GAN Performance Results

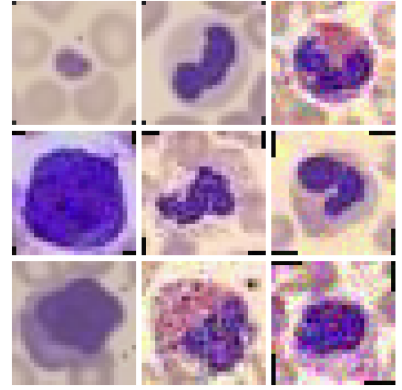
Configuration	Learning Rate	Latent Dim	Feature Maps	Embedding Size	FID Score	Std Dev
Base Model	1e-4	100	128	100	246.33	0.50
Grid Search	1e-4	100	128	100	280.33	-
Final Model	1e-4	100	128	100	146.99	0.83

- **Base Model:** Achieved 246.33 ± 0.50 with conventional architecture
- **Grid Search:** Identified optimal hyperparameters (FID=280.33 in limited 25-epoch training)
- **Final Model:** Achieved best FID= 146.99 ± 0.83 after training with the best configuration of hyperparameters

4.5 Diffusion Model Results



(a) Training loss curve showing convergence



(b) Examples of generated samples

Fig. 6: Diffusion Model Training and Results: (a) Training loss progression demonstrating stable optimization. (b) Generated samples showing the model's ability to produce diverse blood cell images.

Table 8: Diffusion Model Performance Results

Configuration	Base Channels	Attention	FID Score	Std Dev
Base Model	32	No	134.26	1.30
Manual Opt.	64	Yes	113.72	0.77
Grid Search	64	Yes	107.34	-
Final Model	64	Yes	73.43	0.11

- **Base Model:** Achieved FID= 134.26 ± 1.30 with basic U-Net architecture
- **Manual Improvements:** Reduced FID to 113.72 ± 0.77 through architectural enhancements
- **Grid Search:** Identified optimal hyperparameters (FID=107.34 in limited training)
- **Final Model:** Achieved best FID= 73.43 ± 0.11 with full training

4.6 Comparative Analysis

Key observations from the comparative analysis:

- DCGANs achieved the best absolute performance (FID=37.83)

Table 9: Final Model Comparison

Model	Best FID	Training Time	Memory Usage	Stability	Generation Speed
VAE	67.94 ± 0.37	Fast	Low	High	Fast
DAE (Recon.)	9.67 ± 0.05	Fast	Low	High	Fast
DAE (Gen.)	384.32 ± 0.30	Fast	Low	Medium	Fast
DCGAN	37.83 ± 0.49	Fast	Medium	Medium	Fast
CGAN	146.99 ± 0.83	Medium	Medium	Medium	Fast
Diffusion	73.43 ± 0.11	Slow	High	Medium	Slow

- VAEs (FID=67.94) and Diffusion models (FID=73.43) showed competitive performance but the later model required more resources
- DAEs excelled at reconstruction but failed completely at generation
- CGANs provided conditional generation capability at the cost of higher FID (146.99)
- The adversarial models (DCGAN/CGAN) showed the characteristic trade-off between quality and training stability

5 Discussion

5.1 Model-Specific Insights

VAE Performance: The systematic optimization process for the VAE demonstrated the importance of careful hyperparameter tuning. The progression from base model (FID=116.30) to final model (FID=67.94) showed that:

- Moderate latent dimensions (64) outperformed both smaller and larger sizes
- Lower KL weights ($\beta=0.2$) provided better results than stronger regularization
- The combination of MSE and L1 losses helped balance sharpness and diversity

DAE Performance: Confirmed the theoretical expectations:

- Excellent for reconstruction tasks (FID=9.67)
- Completely unsuitable for generation (FID=384.32)

GAN Performance: The GAN variants demonstrated distinct characteristics:

- DCGANs achieved the best raw performance (FID=37.83) but required careful tuning
- CGANs provided class-conditional generation at the cost of higher FID (146.99)
- Both GAN variants showed typical adversarial training challenges
- Spectral normalization and label smoothing proved essential for training stability

Diffusion Model Performance: The diffusion model showed similar improvement patterns:

- Attention mechanisms and improved noise scheduling significantly helped
- The model required more training time and resources than VAE
- Final performance was competitive with VAE but not superior

DCGAN Superiority Despite Instability The DCGAN’s superior performance (FID=37.83) emerges from favorable alignments between its architecture and our task:

- The 28×28 resolution optimally matches DCGAN’s local texture generation strengths
- Spectral normalization and label smoothing effectively stabilized training
- Discrete cell morphologies suit GANs’ implicit modeling better than VAEs’ continuous spaces
- Single-step generation enabled more training iterations than diffusion models

This suggests that for small medical images, well-regularized GANs can outperform theoretically more sophisticated approaches.

5.2 Methodological Considerations

The systematic approach to model development proved valuable:

- Phased development allowed clear attribution of improvements
- Grid search revealed non-intuitive optimal parameters
- Comprehensive evaluation ensured reliable metrics

5.3 Implications for Medical Imaging

Key implications for medical image generation:

- DCGANs are preferable when maximum quality is prioritized and resources allow for extensive tuning
- VAEs offer the best balance for clinical applications needing reliable results
- Diffusion models show promise for high-fidelity generation when computational cost is acceptable
- CGANs provide valuable class control despite quality trade-offs
- DAEs remain useful for specific denoising applications
- Task-specific model selection is crucial

6 Conclusion

These findings provide clear guidance for selecting generative approaches based on clinical requirements, available resources, and desired functionality. The systematic comparison establishes benchmarks for future work in medical image generation.

Our systematic comparison reveals three key lessons for medical image generation:

- Architectural simplicity often outperforms theoretical sophistication for small medical images
- The optimal model depends heavily on specific clinical requirements and constraints
- Standard computer vision metrics may not fully capture medical image quality requirements

Practical Implications The superior performance of DCGANs on this medical imaging task (despite their theoretical instability) suggests that:

- For clinical applications prioritizing image quality above all, well-regularized GAN architectures remain the current best choice
- When training stability is paramount, VAEs offer the most reliable alternative with only modest quality degradation
- Diffusion models may be preferable only when sufficient computational resources are available
- The complete failure of DAEs for generation confirms theoretical limitations in medical imaging contexts

Future Work Directions

- Hybrid approaches combining VAEs’ stability with GANs’ generation quality
- Investigation of more stable GAN architectures specifically for medical imaging
- Improved conditioning mechanisms for CGANs to reduce their quality penalty
- Evaluation on higher-resolution medical imaging datasets
- Clinical validation of synthetic images’ utility for downstream tasks

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